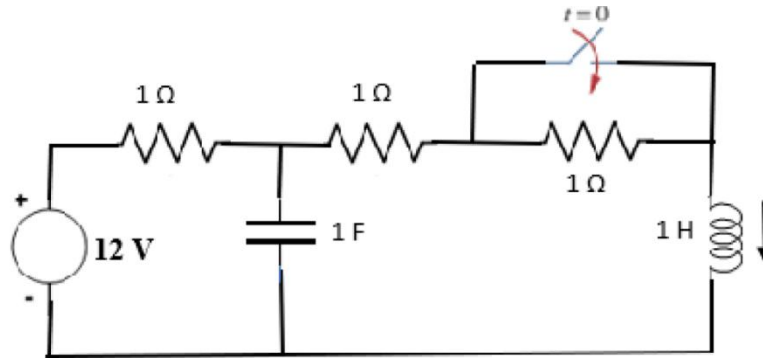


1. (a) V-I characteristics of an electric circuit element is described in the following table. Classify the element into (a) linear and nonlinear (b) unilateral and bilateral (c) active and passive distributed (d) lumped and (e) Deterministic and Stochastic. Justify your answers.

V (volts)	-3	-2	-1	0	1	2	3
I (amps)	9	4	1	0	1	4	9

(b) Prove that the circuit described by the equation $\frac{dy}{dt} + 2y(t) = x(t)$ is linear.

2. The switch shown in figure (b) is closed at $t = 0$. Determine the inductor current for $t > 0$



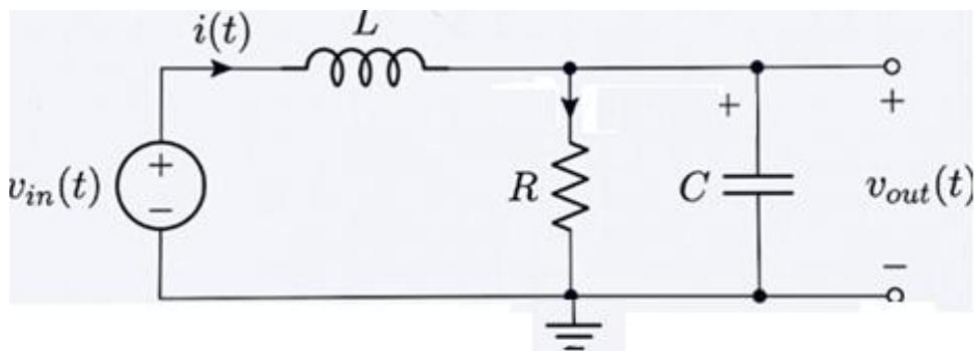
3. Show that, in a graph, all the branch currents can be written in terms smaller number of independent link currents using suitable matrix.

4. The state equations of a second order electrical circuit is given as

$$\begin{aligned}\dot{x}_1 &= -3x_1 + x_2 + u \\ \dot{x}_2 &= -2x_2\end{aligned}\quad y = 2x_1 + x_2$$

Convert the state equations to the canonical form of state equations.

5. A second -order electrical circuit is shown in the figure below, where v_{in} is the input and v_{out} is the output. $L = 1\text{ H}$, $C = 1\text{ F}$, $R = 0.25\text{ }\Omega$. Use the network function approach to find out the input to the circuit when the steady-state output is $v_{out} = 2 + 2\sin 2t + 5\sin 10t\text{ V}$



6. Draw a simulation diagram to solve the following differential equation

$$\ddot{x} + \dot{x} + x = 1$$

Also plot the solution of the equation for $t > 0$, if initial conditions are zero.